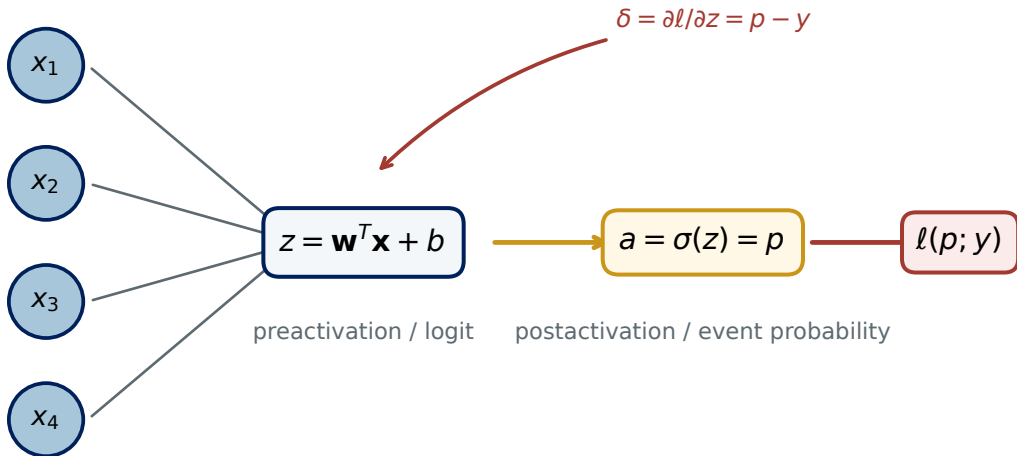


**A**

## One logistic neuron



**Forward: compute a probability. Backward: transmit a score derivative.**

**B**

## Statistical specification

|                         |                           |
|-------------------------|---------------------------|
| <b>Target</b>           | $y \in \{0, 1\}$          |
| <b>Activation</b>       | $\sigma(z)$               |
| <b>Output meaning</b>   | $p = P(Y = 1 \mid X = x)$ |
| <b>Training loss</b>    | Bernoulli log loss        |
| <b>Score derivative</b> | $p - y$                   |
| <b>Decision</b>         | separate threshold policy |

**The affine calculation alone does not define the estimator.**